

1. Создал в папке с `zImage` `vexpress-v2p-ca9.dtb` файл с названием `init.c` с таким содержанием:

```
#include <stdio.h>

#include <unistd.h>

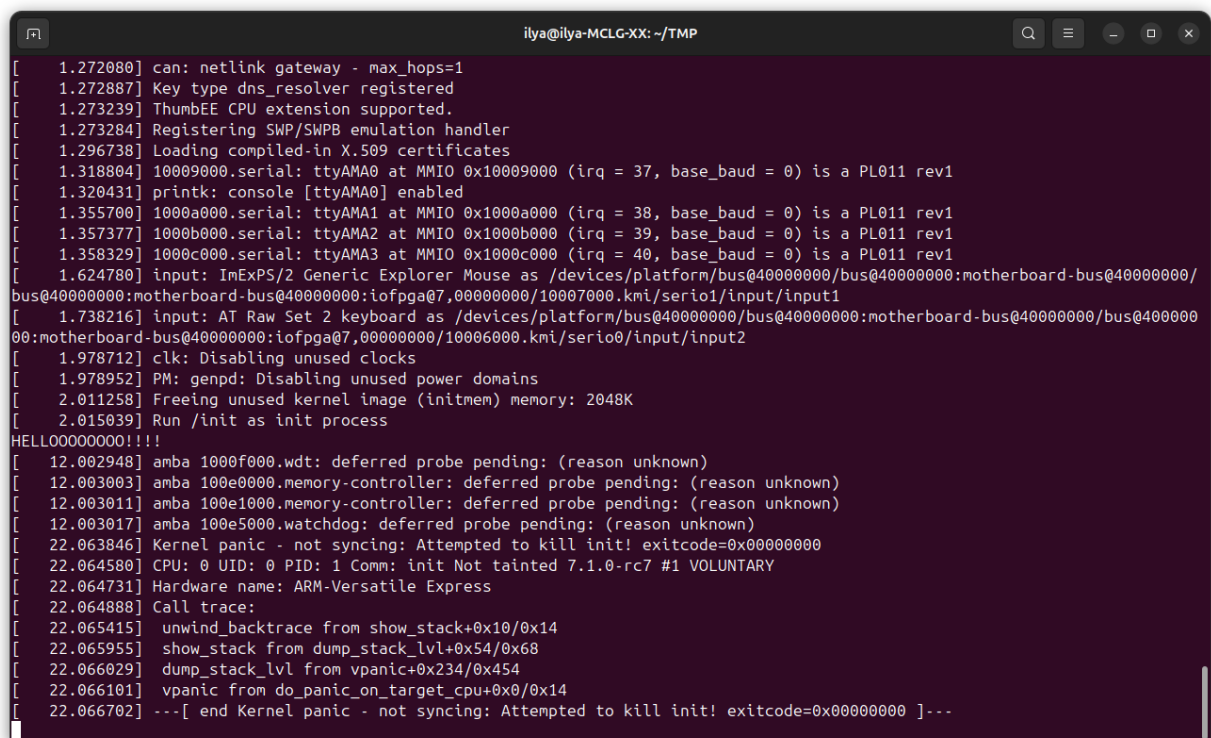
int main(void)
{
    printf("HELLOOOOOOOO!!!!\n");
    sleep(20);
}
```

Компилирую файл командой `arm-linux-gnueabi-gcc -static init.c -o init`

Ввожу команду `echo init | cpio -o -H newc | gzip > initramfs.cpio.gz`

Запускаю QEMU командой `QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic`

Всё успешно запустилось и мы видим ту самую строку `HELLOOOOOOOO!!!!` и через 20 секунд, указанных в `sleep(20)` произошел **Kernel panic**



```
ilya@ilya-MCLG-XX: ~/TMP
[ 1.272880] can: netlink gateway - max_hops=1
[ 1.272887] Key type dns_resolver registered
[ 1.273239] ThumbEE CPU extension supported.
[ 1.273284] Registering SWP/SWPB emulation handler
[ 1.296738] Loading compiled-in X.509 certificates
[ 1.318804] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 1.320431] printk: console [ttyAMA0] enabled
[ 1.355700] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 1.357377] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 1.358329] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 1.624780] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000/motherboard-bus@40000000/
bus@40000000/motherboard-bus@40000000/iopfpga@7,00000000/10007000.kmi/serio1/input/input1
[ 1.738216] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000/motherboard-bus@40000000/bus@400000
00/motherboard-bus@40000000/iopfpga@7,00000000/10006000.kmi/serio0/input/input2
[ 1.978712] clk: Disabling unused clocks
[ 1.978952] PM: genpd: Disabling unused power domains
[ 2.011258] Freeing unused kernel image (initmem) memory: 2048K
[ 2.015039] Run /init as init process
HELLOOOOOOOO!!!!
[ 12.002948] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 12.003003] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 12.003011] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 12.003017] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
[ 22.063846] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[ 22.064580] CPU: 0 UID: 0 PID: 1 Comm: init Not tainted 7.1.0-rc7 #1 VOLUNTARY
[ 22.064731] Hardware name: ARM-Versatile Express
[ 22.064888] Call trace:
[ 22.065415] unwind_backtrace from show_stack+0x10/0x14
[ 22.065955] show_stack from dump_stack_lvl+0x54/0x68
[ 22.066029] dump_stack_lvl from vpanic+0x234/0x454
[ 22.066101] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 22.066702] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---
```

2. Скачал `busybox`

Создаю дефолтный для арма конфиг `ARCH=arm make defconfig`

Запускаем `ARCH=arm make menuconfig`

В `settings` выставляем *** Build static binary**

В `settings` в **Cross compiler prefix** пишем `arm-linux-gnueabi-`

Сохраняем и выходим из настроек

Собираем через **make -j12**

Появилось много ошибок, связанных с **networking/tc.c** и бинарный файл не появился

Чтобы исправить ошибку в настройках конфига в разделе **Networking Utilities** отключаю **tc**

Пишу **make clean** и опять запускаю **make -j12**

Теперь всё завершилось без ошибок.

Командой **file busybox** проверяем всё ли собралось как надо

busybox: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1]=9ae6aec1f3a381b5ff3fd0ca9ec743e21dd12035, for GNU/Linux 3.2.0, stripped

Запускаю **make install**

Появилась папка **_install**, внутри которой находится наша корневая файловая система

Архивируем всё командой **find . | cpio -o -H newc | gzip > initramfs.cpio.gz**

Копируем полученный файл в нашу папку ко всем остальным файлам

Запускаем командой **QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zimage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic**

Всё работает, проверил команду **ls** и создал и смонтировал **proc**, чтобы работала команда **ps**

```
ilya@ilya-MCLG-XX: ~/TM
[ 1.456725] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 1.883217] input: InExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000
0007000.kmi/serio1/input/input2
[ 2.080069] clk: Disabling unused clocks
[ 2.080312] PM: genpd: Disabling unused power domains
[ 2.113122] Freeing unused kernel image (initmem) memory: 2048K
[ 2.115943] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ #
~ #
~ # ls
bin          initramfs.cpio.gz  root          usr
dev          linuxrc           sbin
~ # [ 12.641782] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 12.641836] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 12.641844] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 12.641851] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
~ # mkdir proc
~ # ls
bin          initramfs.cpio.gz  proc          sbin
dev          linuxrc           root          usr
~ # mount -t proc [ 271.551120] random: crng init done
proc proc
~ # mount -t proc proc proc
~ # ps
PID  USER      TIME  COMMAND
  1  0          0:00  /bin/ash
  2  0          0:00  [kthreadd]
  3  0          0:00  [pool_workqueue_]
  4  0          0:00  [kworker/R-rcu_g]
  5  0          0:00  [kworker/R-sync_]
  6  0          0:00  [kworker/R-kvfre]
  7  0          0:00  [kworker/R-slub_]
  8  0          0:00  [kworker/R-netns]
  9  0          0:00  [kworker/0:0-rcu]
 10  0          0:00  [kworker/0:1-eve]
 11  0          0:00  [kworker/0:0H]
 12  0          0:01  [kworker/u16:0-p]
 13  0          0:00  [kworker/R-mm_pe]
 14  0          0:00  [ksoftirqd/0]
 15  0          0:00  [rcu_sched]
 16  0          0:00  [rcu_exp_par_gp_]
 17  0          0:00  [rcu_exp_gp_kthr]
 18  0          0:00  [migration/0]
 19  0          0:00  [cpuhp/0]
 20  0          0:00  [cpuhp/1]
 21  0          0:00  [migration/1]
 22  0          0:00  [ksoftirqd/1]
 23  0          0:00  [kworker/1:0]
 24  0          0:00  [kworker/1:0H]
```